

ASSIGNMENT 08  
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ROLL NO - CSB24021

```
import React, { useState } from "react";

export default function EnrollmentDashboard() {

  const [students, setStudents] = useState(new Map());

  // Add new student
  const addStudent = (student) => {
    setStudents(prev => {
      const newMap = new Map(prev);
      newMap.set(student.id, student);
      return newMap;
    });
  };

  // Remove student by ID
  const removeStudent = (id) => {
    setStudents(prev => {
      const newMap = new Map(prev);
      newMap.delete(id);
      return newMap;
    });
  };

  // Students sorted by GPA (descending)
  const sortedStudents = [...students.values()]
    .sort((a, b) => b.gpa - a.gpa);

  // Unique courses
  const uniqueCourses = [...students.values()].reduce((acc, s) => {
    s.enrolledCourses.forEach(c => acc.add(c));
    return acc;
  }, new Set());

  // Filter students by course
  const filterByCourse = (course) =>
    [...students.values()].filter(s =>
      s.enrolledCourses.has(course)
    );

  return (
    <div>
      <h2>Enrolled Students</h2>
    </div>
  );
}
```

```
{sortedStudents.map(s => (  
  <div key={s.id}>  
    <p>{s.name} - GPA: {s.gpa}</p>  
    <p>Courses: {[...s.enrolledCourses].join(", ")}</p>  
    <button onClick={() => removeStudent(s.id)}>Remove</button>  
  </div>  
))}  
  
  <h3>All Unique Courses</h3>  
  {[...uniqueCourses].map(c => (  
    <p key={c}>{c}</p>  
  ))}  
</div>  
);  
}
```